Maryam Aminu Mukhtar

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EDUCATION

Northeastern University, Boston, MA December 2025

Master of Science, Wireless and Network Engineering

Courses: Fundamentals of Computer Networks, Wireless Sensor Networks and the Internet of Things, Mobile and Wireless Networking

University of Massachusetts Boston, Boston, MA May 2023

Bachelor of Science, Computer Engineering (magna cum laude)

- Awarded Engineering Department's Outstanding Achievement in Engineering Award for exceptional academic and research work

TECHNICAL SKILLS

- **Wireless & RF:** wireless sensor networks, RF protocols (IEEE 802.15.4, Zigbee), Wi-Fi, LTE, 5G NR (signaling, link adaptation), TDD, BWP, real-time monitoring, wireless troubleshooting
- **Network & Systems:** OSI model, TCP/IP, UDP, DNS, DHCP, LAN/WAN, IP addressing & subnetting, routing protocols (OSPF, BGP), network troubleshooting (iperf), tools (Wireshark, ns-3)
- **Languages:** Python (Pandas, NumPy, Matplotlib), C/C++, MATLAB, bash
- **Tools & Platforms:** Linux/Ubuntu, Git/GitHub, Docker, GNU Radio, Salesforce, NetCloud, Ignition SCADA, Microsoft Office Suite

WORK EXPERIENCE

Sullivan and McLaughlin, Dorchester, MA

Network Operations Co-op May 2024 – December 2024

- Diagnosed and resolved network connectivity failures across 100+ routers using **NetCloud** management platform, performing remote resets, modem diagnostics, and fault isolation to determine hardware vs. configuration root causes
- Processed and configured Cradlepoint LTE/5G routers including SIM coordination, **NetCloud** enrollment, and connectivity validation for enterprise client orders using **Salesforce**
- Coordinated directly with carriers and vendors (Cradlepoint) on the **RMA process** for unrecoverable devices and on SIM/activation issues, managing replacements to minimize client disruption

Institute for Intelligent Networked Systems, Northeastern University, Boston, MA

Lab Research Assistant Sept 2023 – April 2024

- Deployed and managed **Docker** containers for **GNU Radio** and web-server environments, streamlining setup and improving reproducibility across research systems
- Validated, analyzed, and organized sensor datasets in **Python (NumPy, Pandas)** for a Parkinson's health-monitoring project, verifying data quality and correctness, preparing it for downstream use
- Built a web server with real-time visualization for a spectrum-sensing project, making the captured spectrum directly observable to confirm the system was reading data as expected

PROJECT EXPERIENCE

University of Massachusetts Boston, Boston, MA

Augmented Reality Informational System (ARIS) for Astronauts, Capstone project Sept 2022 – May 2023

- Led a multidisciplinary team to design and build an **end-to-end IoT system** for real-time health monitoring, designing the overall system architecture and integration across sensing, wireless communication, and data-display layers
- Designed and implemented **full-duplex RF communication (XBee/Zigbee, IEEE 802.15.4)** between distributed hubs, configuring the modules and running range tests measuring accuracy, packet loss, latency, and RSSI out to ~400 ft
- Integrated distributed sensors, **edge computing (Raspberry Pi)**, and a **real-time data** pipeline delivering sensor data to a monitoring interface
- Built a **Python socket server/client** architecture to stream sensor data reliably from a Raspberry Pi to the central monitoring system

Northeastern University, Boston, MA

5G Network Performance Analysis and Optimization, EECE 7364 Lab Series Feb 2025 – April 2025

- Configured and operated a live **5G NR testbed (OAIBOX)** end to end, setting up the UE (Quectel modem via AT commands), gNB, and 5G core, including band locking, bandwidth/BWP, and TDD slot configurations
- Captured and analyzed control- and user-plane signaling (NAS, RRC, NGAP) using **Wireshark** and OAI logs to trace the UE attachment and registration procedure across the UE, gNB, and core network
- Measured and analyzed throughput, latency, and signal quality (RSRP/RSRQ/SINR/BLER/MCS) with **iperf** across bandwidth, distance, LoS/NLoS, and modulation scenarios, correlating radio conditions to link-adaptation behavior

LEADERSHIP & INVOLVEMENT

Paper Trails & Tales Book Club, Founder & Creative Director, Boston, MA May 2024 – Present

- Founded and lead a book club, planning and running 26+ events end to end, including venue selection, scheduling, and member communications
- Grew it to 16 active members and 64+ unique attendees, sustaining consistent turnout of 7–11 per event